

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

April 2013

EC207 Control Systems

Date: 24.04.2013, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

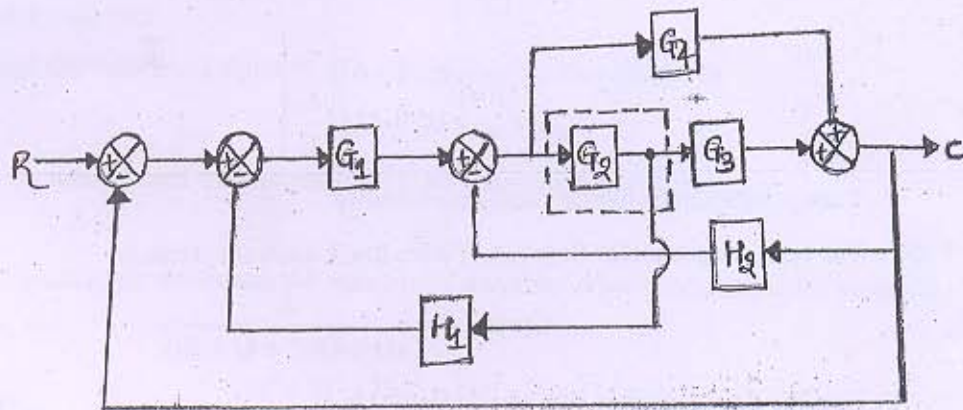
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Use Graph Papers.

SECTION - I

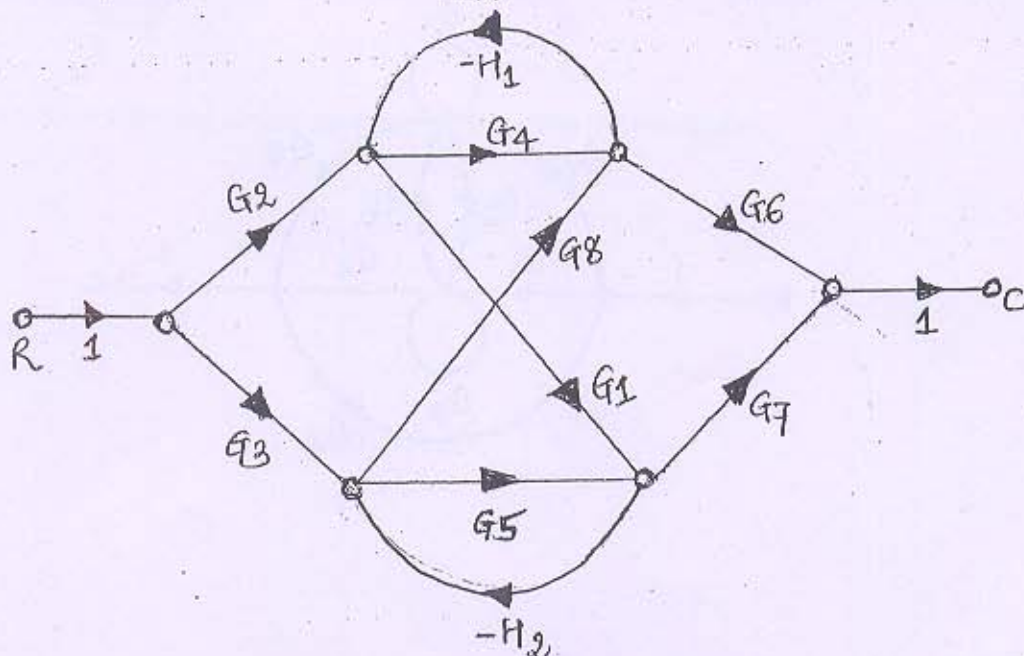
Q - 1 (a) Give the differences between block diagram representation and signal flow graph representation of system. [04]

(b) Discuss the difference between open loop control system and close loop control system. [03]

Q - 2 (a) Determine $\frac{C(s)}{R(s)}$ system shown below using block diagram reduction techniques. [07]

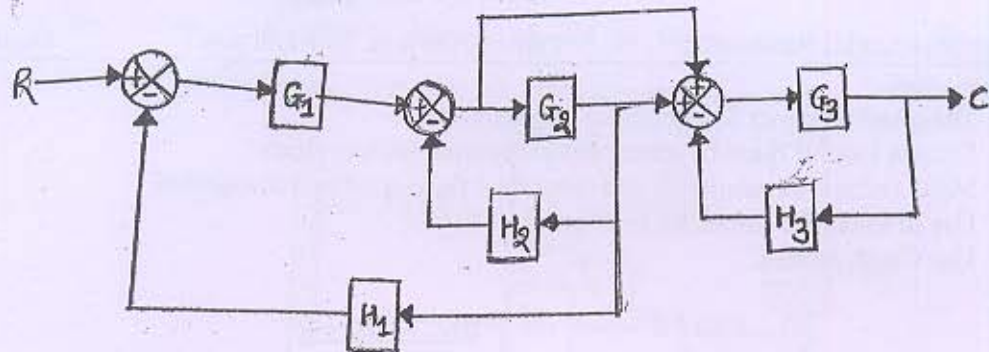


(b) Using Mason's gain formula, find $\frac{C(s)}{R(s)}$ of the signal flow graph shown in figure. [07]



OR

- Q - 2 (a) Evaluate the transfer function $\frac{C(s)}{R(s)}$ for the system shown in figure below, using block diagram reduction techniques. [08]



- (b) A system is described by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & -1 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

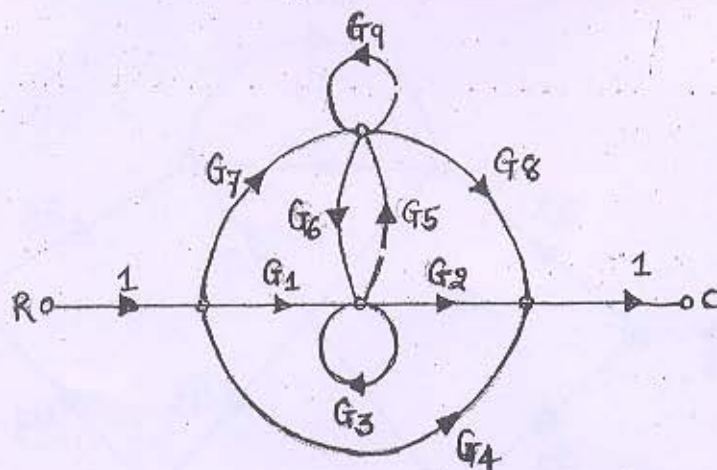
Test its state controllability and observability.

- Q - 3 (a) The open-loop transfer function of a feedback control system is

$$G(s)H(s) = \frac{K}{s(s+4)(s^2+4s+20)}$$

Find the Root Locus as K is varied from 0 to ∞ .

- (b) Using Mason's gain formula, find $\frac{C}{R}$ of the signal flow graph shown in figure. [07]



OR

- Q - 3 (a) Consider a unity feedback control system with an open-loop transfer function [07]

$$G(s) = \frac{K(s+1)(s+2)}{(s+0.1)(s-1)}$$

Draw the Root Locus of the system with the gain K as a variable.

- (b) Write short note on Synchro transmitter and receiver. [07]

SECTION - II

- Q - 4 (a) Sketch the asymptotic bode plot for the transfer function [07]

$$G(s)H(s) = \frac{10(s+5)}{s(s+0.5)(s+1)}$$

Comment on Stability.

- Q - 5 (a) A unity feedback control system has [07]

$$G(s)H(s) = \frac{80}{s(s+2)(s+20)}$$

Draw the bode plot.

- (b) Consider characteristics equation of a closed-loop control system as [07]

$$s^3 + 3Ks^2 + (K+2)s + 4 = 0$$

Check its stability and find the range of K so that system is stable.

OR

- Q - 5 (a) Using Routh's test determine the stability of a system whose characteristics equation [06]
is given by

$$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$$

- (b) Using Nyquist criterion, examine the closed-loop stability of a system whose open- [08]
loop transfer function is

$$G(s)H(s) = \frac{50}{(s+1)(s+2)}$$

- Q - 6 (a) Prove the following torque equations for the shafts of Gear Trains. [06]

$$J_{1eq} \ddot{\theta}_1 + f_{1eq} \dot{\theta}_1 + \left(\frac{N_1}{N_2} \right) T_L = T_M$$

And

$$J_{2eq} \ddot{\theta}_2 + f_{2eq} \dot{\theta}_2 + T_L = \left(\frac{N_2}{N_1} \right) T_M$$

(b) Sketch the Polar plot for a system with

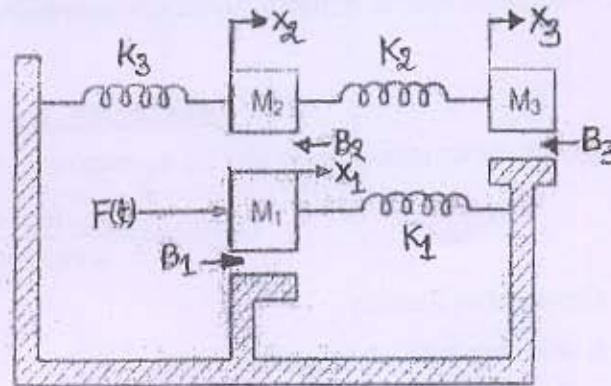
[08]

$$G(s)H(s) = \frac{10}{s(s+1)(s+2)}$$

Calculate its gain margin in dB and comment on stability.

OR

Q - 6 (a) For the following translation system shown in Figure, Obtain the differential equations and draw the electrical network using force-voltage analogy. [08]



(b) A unity feedback system has open loop transfer function given below

[06]

$$G(s) = \frac{(1+0.4s)}{s(s+0.6)}$$

- 1) Obtain characteristic equation.
- 2) Obtain unit step response.
- 3) What is rise time?
- 4) What is peak overshoot?
- 5) Find out settling time for the system.
